MediStock:

1. Title

MediStock: Medical Inventory Management Platform

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2. Objective

Build a full-stack web application (React.js frontend + Spring Boot backend) that allows pharmacies, hospitals, and healthcare organizations to manage medicine inventory, track stock availability, monitor expiry dates, maintain supplier records, and generate inventory analytics.

The system provides real-time stock updates, low-stock notifications, expiry monitoring, supplier management, and inventory dashboards for efficient medicine inventory management.

Outcomes

- Developed and deployed a full-stack React.js + Spring Boot application.
- Implemented secure user authentication using JWT and OAuth2 login.
- Built medicine inventory and stock management functionality.
- Implemented expiry date tracking and low-stock alerts.
- Added supplier and purchase management systems.
- Created dashboards for inventory analytics and stock monitoring.
- Implemented medicine search, filtering, and category management.
- Deployed frontend and backend using PostgreSQL in production.
- Added notification systems for expiry and inventory alerts.

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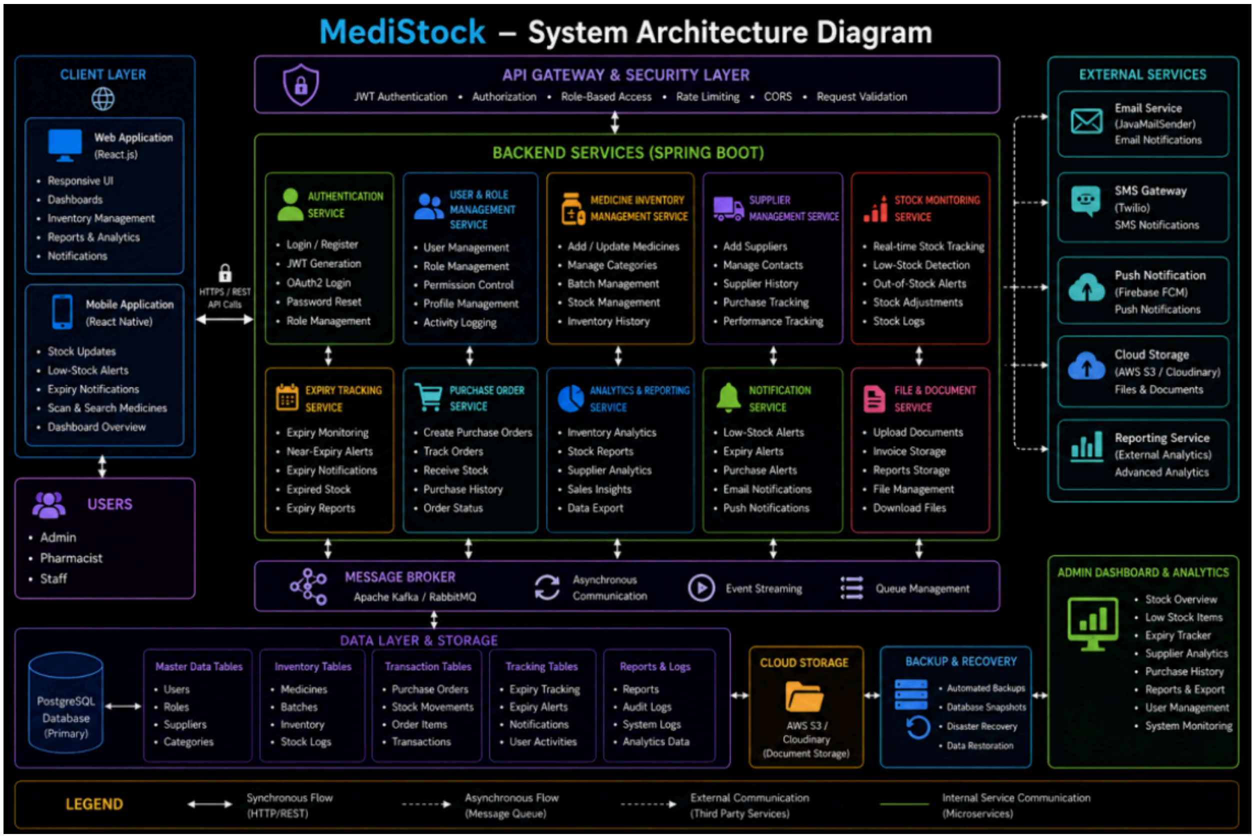
3. Architecture Diagram



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4. Modules to be Implemented

1. User Authentication & Role-Based Access

- JWT Authentication
- OAuth2 Login
- Role Management
- Password Reset
- User Profile Management

Roles

- Admin
- Pharmacist
- Staff

2. Medicine Inventory Management

- Add medicines
- Update stock quantities
- Delete medicines
- Manage medicine categories
- Track medicine batches
- Inventory history tracking

Medicine Information

- Medicine Name
- Batch Number
- Category
- Supplier
- Quantity
- Manufacturing Date
- Expiry Date
- Price

3. Supplier Management System

- Add supplier details
- Manage supplier contacts
- Track supplier purchases
- Supplier history management
- Supplier performance tracking

Supplier Information

- Supplier Name
- Contact Number

- Email
- Address
- Supplied Medicines

4. Stock Monitoring & Alerts

- Real-time stock tracking
- Low-stock alerts
- Out-of-stock notifications
- Automatic stock updates
- Inventory movement tracking

5. Expiry Tracking System

- Medicine expiry monitoring
- Expiry notifications
- Near-expiry medicine tracking
- Expired stock management
- Expiry reports

6. Search & Filtering System

Users can search medicines by:

- Name
- Category
- Supplier
- Batch Number
- Expiry Date
- Stock Status

7. Dashboard & Analytics

Pharmacist Dashboard

- Inventory overview
- Low-stock items
- Expiring medicines
- Purchase summary
- Supplier insights

Admin Dashboard

- Inventory analytics
- User activity
- Supplier analytics
- Stock movement reports
- System monitoring

8. Notification & Reminder System

- Low-stock notifications
- Expiry alerts
- Inventory reminders
- Purchase alerts
- Email notifications
- Push notifications

9. Reports & Data Export

- Generate inventory reports
- Download PDF reports
- Download Excel reports
- Export stock data
- Purchase history reports

- Expiry reports

10. Final Integration, Testing & Deployment

- Frontend and backend integration
- API validation and testing
- End-to-end workflow testing
- Security testing
- Performance optimization
- Docker containerization
- Production deployment
- Monitoring and logging setup
- Documentation and user guides

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5. Week-wise Module Implementation and High-Level Requirements

Milestone 1: Week 1 & 2 — Requirements, Database Design & Backend Setup

Tasks

- Define project scope and user roles.
- Design database schema:
    - Users
    - Roles
    - Medicines
    - Suppliers
    - Inventory
    - Stock Logs

    - Purchase Orders
    - Expiry Tracking
    - Notifications
    - Reports
- Initialize Spring Boot backend.
- Configure PostgreSQL / MySQL.
- Implement JWT authentication APIs.
- Create React frontend skeleton.

Outcomes

- Backend and frontend architecture setup completed.
- Authentication flow functional.
- Database schema finalized.
- Role-based access configured.

Milestone 2: Week 3 & 4 — Inventory & Supplier Management

Tasks

- Implement medicine inventory APIs.
- Build supplier management system.
- Create medicine dashboard pages.
- Implement stock tracking functionality.
- Add medicine search and filtering.

Outcomes

- Inventory management operational.
- Supplier workflow functional.
- Dynamic stock tracking implemented.

Milestone 3: Week 5 & 6 — Expiry Tracking & Notifications

Tasks

- Implement expiry tracking system.
- Build low-stock alert functionality.
- Add notification integrations.
- Generate analytics APIs.
- Implement report generation system.

Outcomes

- Expiry monitoring operational.
- Alert and notification system functional.
- Inventory analytics completed.

Milestone 4: Week 7 & 8 — Analytics, Testing & Deployment

Tasks

- Build analytics dashboard.
- Add charts and reporting visualizations.
- Implement testing and validations.
- Deploy frontend/backend using AWS, Render, or Railway.
- Configure environment variables, SSL, and CORS.

Outcomes

- Fully deployed production-ready application.
- Inventory monitoring systems operational.
- Complete end-to-end inventory workflow demonstrable.

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6. Evaluation Criteria

Milestone 1 (Week 2)

- Spring Boot backend scaffolding complete.
- JWT authentication implemented.
- Database schema finalized.
- Frontend authentication flow working.

Milestone 2 (Week 4)

- Medicine inventory management operational.
- Supplier management functioning.
- Search and filtering system implemented.

Milestone 3 (Week 6)

- Expiry tracking operational.
- Notification system functional.
- Inventory analytics workflow completed.

Milestone 4 (Week 8)

- Fully deployed frontend and backend.
- Dashboard and reporting systems operational.
- Demonstrable end-to-end inventory management workflow.

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7. Tools & Tech Stack

Programming Language

Backend

- Java
- Spring Boot

Frontend

- JavaScript
- React.js

Database

Local Development

- MySQL

Production

- PostgreSQL

Authentication

- Spring Security
- JWT Authentication
- OAuth2 (Google Login)

Libraries & Frameworks

Backend

- Spring Boot
- Spring Security
- Spring Data JPA
- Hibernate
- Maven

Frontend

- React
- React Router
- Axios
- Tailwind CSS

- Context API
- Framer Motion

Notifications

- Firebase Cloud Messaging (FCM)
- Twilio
- JavaMailSender

Testing

- JUnit
- Mockito
- Postman
- React Testing Library

Dev & Deployment Tools

- IntelliJ IDEA
- VS Code
- Git & GitHub
- Docker
- Docker Compose
- GitHub Actions
- AWS / Render / Railway
- Postman

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8. Performance Metrics

Inventory Management Metrics

- Stock tracking accuracy

- Inventory update success rate
- Inventory reconciliation accuracy
- Medicine search efficiency

Expiry & Alert Metrics

- Expiry detection accuracy
- Low-stock alert accuracy
- Notification delivery success rate
- Expired stock identification rate

Supplier Management Metrics

- Supplier performance tracking accuracy
- Purchase order processing efficiency
- Supplier response monitoring

System Performance Metrics

- API response time
- Dashboard loading speed
- Concurrent user handling capacity
- Database query performance

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9. Example Quantitative Goals

Inventory Tracking

Maintain accurate real-time medicine inventory and stock movement records.

Expiry Monitoring

Identify and notify users of near-expiry and expired medicines proactively.

Supplier Management

Track supplier performance and purchase activities efficiently.

Inventory Analytics

Provide actionable insights through inventory dashboards and reports.

Platform Performance

Support pharmacies, hospitals, and clinics with stable and scalable inventory management workflows.